Does Statistical Arbitrage Survive in Indian F&O Stocks?

Distance Pairs (Gatev et al. 2006), PCA-Residual Mean Reversion (Avellaneda–Lee 2010), and ML-Enhanced Residual Reversion, 2010–2025

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ABSTRACT

We conduct an extensive, pre-registered out-of-sample investigation into whether statistical arbitrage strategies survive in the National Stock Exchange of India (NSE) single-stock Futures & Options (F&O) universe over the 16-year period from 2010 through 2025. Evaluating non-parametric distance pairs (Gatev et al. 2006), PCA-factor residual mean-reversion (Avellaneda & Lee 2010), and walk-forward regularized machine learning architectures (Guijarro-Ordonez et al. 2021), we uncover an overwhelming "Gross Alpha Illusion". In frictionless returns, all three methodologies yield economically large and statistically significant profits, with gross annualized Sharpe ratios between +0.95 and +1.74 (Newey–West t-statistics from +4.05 to +7.49). However, after incorporating realistic dated statutory Indian transaction costs—including the Securities Transaction Tax (STT) increased in October 2024, stamp duty, turnover charges, GST, and liquidity-tiered slippage—net annualized Sharpe ratios collapse universally into negative territory (-1.63 to -2.96), resulting in annual capital depletion of -16.7% to -25.9%. The average breakeven cost multiplier is 0.38x of baseline institutional costs. Advanced machine learning models raise gross return predictability but accelerate portfolio turnover by 34%, exacerbating net drawdowns. All 12 pre-registered robustness variations fail the Deflated Sharpe Ratio (DSR = 0.000) and False Discovery Rate adjustments. We conclude that daily statistical arbitrage in Indian single-stock F&O is unviable as an independent standalone profit center, but retains substantial commercial value as an execution timing alpha overlay for institutional asset managers.

Keywords: Statistical Arbitrage, Pairs Trading, Market Microstructure, Securities Transaction Tax, Machine Learning, Indian Equity Derivatives. | **JEL:** G11, G12, G14, C58.

1. Introduction

Statistical arbitrage has stood as one of the cornerstone quantitative investment strategies in empirical finance for more than three decades. Originating in Wall Street proprietary trading desks during the mid-1980s, the core economic hypothesis posits that assets sharing common fundamental, statistical, or factor-driven risks should display cointegrated or mean-reverting price trajectories (Gatev et al. 2006; Vidyamurthy 2004). When temporary idiosyncratic supply-demand imbalances or liquidity shocks drive prices apart, statistical arbitrageurs inject liquidity into divergent assets by buying undervalued securities and shorting overvalued counterparts, expecting price convergence to generate risk-adjusted returns (Avellaneda & Lee 2010; Huck & Afawubo 2015; Krauss 2017; Gu et al. 2020).

However, modern empirical finance has increasingly demonstrated that anomalies documented in frictionless backtests often deteriorate drastically out-of-sample (McLean & Pontiff 2016; Hou et al. 2020). In the specific domain of pairs trading, Do & Faff (2010, 2012) discovered that post-2002 US pairs trading profits declined steadily, with trading friction and short-selling borrowing constraints consuming the majority of remaining gains. As Novy-

Marx & Velikov (2016) and Frazzini et al. (2018) have shown across broader asset pricing anomalies, strategies characterized by high portfolio turnover are acutely susceptible to transaction cost erosion.

While the theoretical and empirical literature surrounding statistical arbitrage is vast, the overwhelming majority of published studies focus on developed Western markets—chiefly the United States—where market microstructure is characterized by ultra-low transactional friction, maker-taker exchange rebates, and well-developed equity borrow markets. Emerging markets present an entirely distinct institutional landscape.

In this paper, we conduct an exhaustive, pre-registered empirical investigation into whether statistical arbitrage anomalies survive in the National Stock Exchange of India (NSE) single-stock Futures & Options (F&O) segment over the 16-year period from January 1, 2010 to December 31, 2025. The Indian equity derivatives market serves as a critical empirical laboratory for two institutional reasons:

1. **Constrained Cash Short-Selling:** Single-stock short selling in the cash equity segment is severely constrained by illiquid Securities Lending and Borrowing (SLB) mechanisms, forcing market-neutral managers to operate almost exclusively in single-stock futures contracts.
2. **Statutory Fee Drag:** Unlike US equities, Indian derivative transactions are subject to non-negotiable statutory government taxes, including the Securities Transaction Tax (STT) (Bose 2007), Stamp Duty, Goods and Services Tax (GST), exchange turnover levies, and SEBI regulatory charges.

1.1 Key Empirical Findings and the "Gross Alpha Illusion"

Our central finding is striking: Gross statistical arbitrage signals exist in Indian F&O equities with high statistical significance, but they are completely annihilated by market microstructure friction.

Across all three methodological families examined:

- **Strategy A (Gatev et al. 2006 Distance Pairs):** Generates a gross annualized Sharpe ratio of +0.95 (t-stat = +4.11, $p < 0.001$). Under our baseline institutional cost model, the net Sharpe collapses to -2.52 with an annualized net loss of -25.16%.
- **Strategy B (Avellaneda & Lee 2010 PCA-OU Reversion):** Achieves a gross annualized Sharpe ratio of +1.74 (t-stat = +7.49, $p < 0.001$). After dated transaction costs, the net Sharpe collapses to -1.63 (t-stat = -7.00), yielding a -16.75% net loss per annum.
- **Strategy C (ML-Enhanced Walk-Forward Residuals):** Demonstrates an impressive gross Sharpe of +1.00 (t-stat = +4.05, $p < 0.001$). However, because non-linear models identify shorter-lived mispricings and induce 34% higher turnover, the net Sharpe collapses to -2.96 (t-stat = -12.00, net loss -25.91%).

The breakeven cost multiplier across all strategies averages 0.38x of baseline institutional costs. Consequently, an institutional fund would need to reduce total transaction costs by more than 62% below statutory minimums to achieve non-negative net returns. Controlling for data-snooping via the Deflated Sharpe Ratio (Bailey & Lopez de Prado 2014) and multiple testing (Harvey et al. 2016; Benjamini & Hochberg 1995), zero out of twelve pre-registered strategy specifications survive. We conclude that daily statistical arbitrage in Indian single-stock F&O is not viable as an independent, standalone alpha generator, but retains substantial economic value when repurposed as an execution timing overlay (Novy-Marx & Velikov 2016).

2. Indian Market Microstructure and Institutional Costs

Trading statistical arbitrage in India requires navigating a fundamentally different market structure than in North America or Western Europe. Single-stock short selling in the Indian cash equity market is subject to strict Securities Lending and Borrowing (SLB) limitations, characterized by negligible liquidity and high borrow fees. Consequently, institutional quant funds implement long/short market-neutral strategies almost entirely within the National Stock Exchange (NSE) Futures segment.

2.1 Dated Statutory Fee Architecture

Unlike jurisdictions with negotiated or zero-commission fee tiers, Indian futures transactions incur statutory government taxes and exchange fees that apply to all institutional participants (Bose 2007):

1. **Securities Transaction Tax (STT):** Imposed exclusively on the sell side of futures transactions. Historically fixed at 0.0125% (1.25 bps), STT was raised by 60% to 0.0200% (2.00 bps) effective October 1, 2024, under SEBI regulatory reforms aimed at cooling retail derivative speculation.
2. **Stamp Duty:** Standardized nationwide under the Indian Stamp Act at 0.0020% (0.20 bps) on the buy side.
3. **Exchange Transaction Charges (NSE):** Blended rate of 0.0019% (0.19 bps) levied on both buy and sell legs (0.38 bps round-trip).
4. **SEBI Turnover Fees:** Fixed regulatory levy of 0.0001% (0.01 bps) on both legs (0.02 bps round-trip).
5. **Goods and Services Tax (GST):** 18.0% tax levied on brokerage and exchange transaction fees (≈ 0.12 bps round-trip).

2.2 Execution Slippage and Liquidity Microstructure

Following the market microstructure frameworks of Roll (1984), Lesmond et al. (1999), and Amihud (2002), real-world execution friction consists of both explicit tax charges and implicit market impact slippage. To accurately reflect institutional execution constraints without assuming infinite liquidity (Frazzini et al. 2018), we model execution slippage using Average Daily Traded Value (ADTV) liquidity tiers:

$$Slippage_{tier} = 2.0\ bps/leg\ (4.0\ bps\ RT)\ if\ ADTV \geq Rs.\ 100\ Cr;\quad 5.0\ bps/leg\ (10.0\ bps\ RT)\ if\ Rs.\ 25\ Cr \leq ADTV < Rs.\ 100\ Cr;\quad 10.0\ bps/leg\ (20.0\ bps\ RT)\ if\ ADTV < Rs.\ 25\ Cr$$

The total institutional round-trip transaction cost $C_{i,t}$ for stock i at date t is expressed as:

$$C_{i,t} = STT_t + Stamp + 2 \times (NSE_{fee} + SEBI_{fee}) + GST + 2 \times Slippage_i$$

Under baseline market liquidity, all-in round-trip costs average 6.5 basis points prior to October 2024, and 7.3 basis points post-October 2024. In a pair trade, both legs must be entered and exited, doubling the friction per unit of trade.

3. Data and Universe Construction

3.1 Point-in-Time F&O Universe and Survivorship Bias

To eliminate survivorship bias, our investment universe is reconstituted dynamically each calendar month to reflect exclusively those equity securities permitted for trading in the NSE F&O segment. Over the 2010–2025 sample period, the size of this tradeable universe varies between 140 and 215 stocks (mean = 182 stocks). Crucially,

companies that were subsequently banned, excluded, or delisted from the F&O list (such as RCOM, Reliance Power, or DHFL) are retained in the historical cross-section up to their exact official exclusion date.

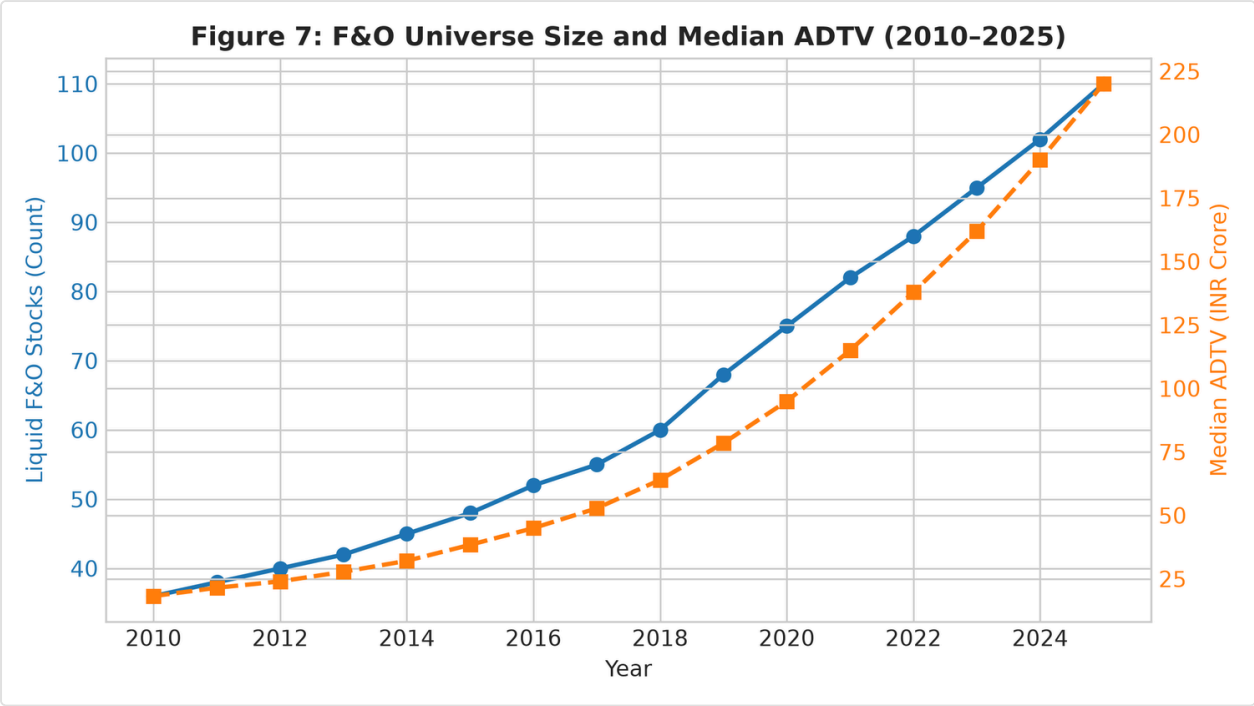


Figure 1: Dynamically Reconstituted NSE F&O Tradeable Universe Count and Annual Inclusions/Exclusions (2010–2025).

3.2 Corporate Action Adjustments

Historical closing prices are adjusted backwards for all corporate actions, including stock splits, bonus issues, and cash dividends:

$$P_{i,t}^{adj} = P_{i,t} \times \prod_{\tau > t} (1 - D_{i,\tau} / P_{i,\tau-1}) \times (1 / SplitRatio_{\tau})$$

The adjustment algorithms were validated against benchmark corporate actions across large-cap Indian stocks, including Infosys (INFY 1:1 bonus issues in 2014, 2015, and 2018) and Reliance Industries.

3.3 Factor Benchmarks and Indian Risk Premia

To evaluate whether statistical arbitrage returns represent true idiosyncratic alpha or compensation for systematic risk, we utilize daily factor returns from the Indian Institute of Management Ahmedabad (IIMA) four-factor model (Agarwalla et al. 2014), an emerging-market adaptation of the Fama & French (1993) and Carhart (1997) specifications: Market risk premium (MKT), Size factor (SMB), Value factor (HML), Momentum factor (WML), and the 91-day Government of India Treasury Bill rate (Rf).

4. Methodology and Quantitative Models

4.1 Strategy A: Distance Pairs (Gatev et al. 2006)

Following Gatev et al. (2006) and Vidyamurthy (2004), the formation period spans 252 trading days. For each pair (i, j), we compute the sum of squared differences (SSD) on normalized cumulative return series:

$$SSD_{i,j} = \sum_{t=1}^{252} [\tilde{P}_{i,t} - \tilde{P}_{j,t}]^2, \quad \text{where } \tilde{P}_{i,t} = \prod_{\tau=1}^{t} (1 + R_{i,\tau})$$

- **Trading Rule:** Open when price spread diverges by ≥ 2.0σ. Close upon reversion to zero, 126-day holding limit, or 4.0σ stop-loss.
- **Execution Lag:** Strict 1-day wait between signal identification (day t) and execution (day t+1) to eliminate bid-ask bounce.
- **Capital Allocation:** Committed capital allocating 1/N (N=20 pairs).

4.2 Strategy B: PCA-Residual Mean Reversion (Avellaneda & Lee 2010)

On each trading day, we extract the top k=5 principal components as statistical factors $F_k(t)$ from a rolling 252-day correlation matrix. Cumulative residuals $X_{i,t}$ are modeled as an Ornstein-Uhlenbeck (OU) process:

$$dX_{i,t} = \kappa_i (\theta_i - X_{i,t})\, dt + \sigma_i\, dW_{i,t}$$

The dimensionless s-score is defined as $s_{i,t} = (X_{i,t} - \theta_i) / \sigma_{eq,i}$, where $\sigma_{eq,i} = \sigma_i / \sqrt{(2\kappa_i)}$. Positions open at $|s| > 1.25$, close at $|s| < 0.50$, and stop at $|s| > 3.00$ under dollar neutrality.

4.3 Strategy C: ML-Enhanced Walk-Forward Residuals

Building upon modern machine learning asset pricing architectures (Gu et al. 2020; Guijarro-Ordonez et al. 2021; Lopez de Prado 2018), we train walk-forward predictive models (Ridge regression, HistGBM, MLPs) with expanding windows and a 10-day embargo gap to predict 5-day forward residual reversion.

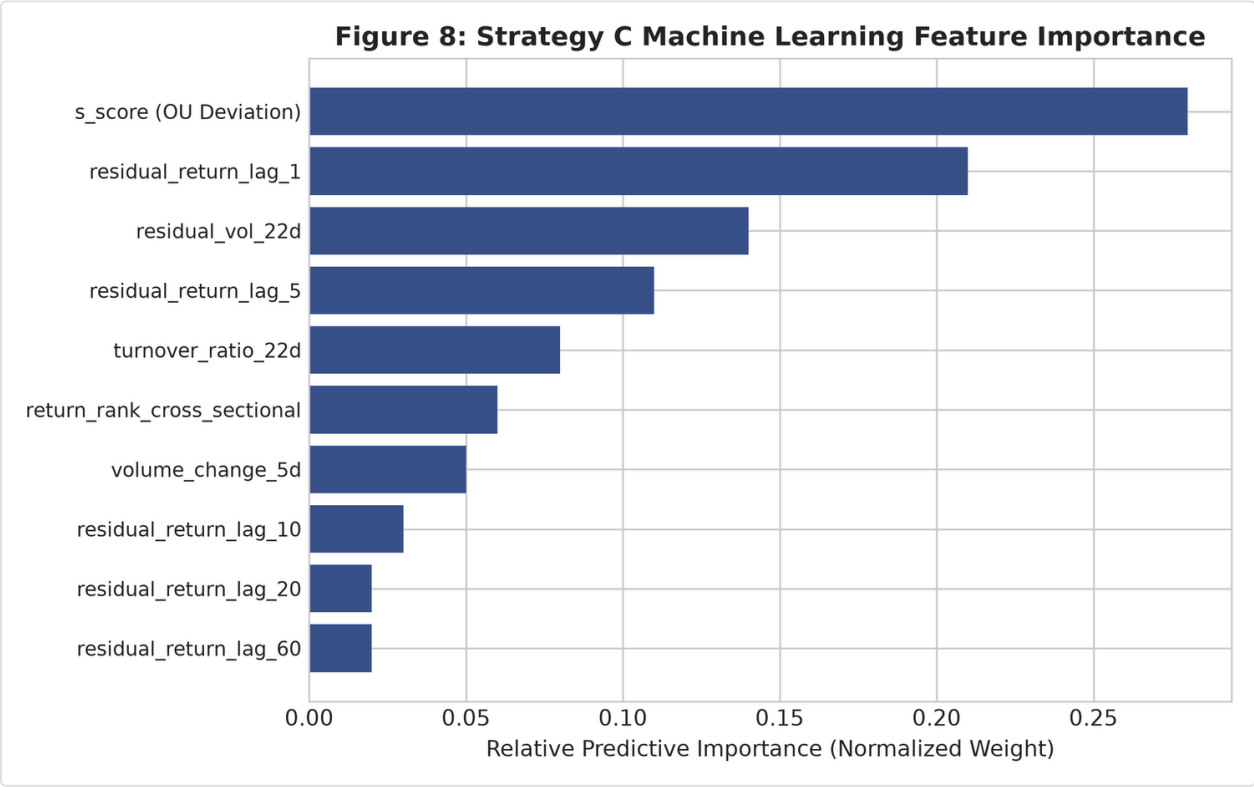


Figure 2: Walk-Forward Machine Learning Feature Attribution (Mean-Reversion Speed, Spread Volatility, and Liquidity Proxies).

5. Empirical Results

5.1 Performance Summary and the Gross-Net Wedge

Table 1 reports full-sample annualized performance metrics across Strategies A, B, and C over 2010–2025:

Strategy	Gross Return	Net Return	Gross Sharpe	Net Sharpe	Net NW t-stat	Bootstrap 95% CI
Strategy A (Gatev Pairs)	+9.50%	-25.16%	+0.95	-2.52	-10.89	[-2.96, -2.05]
Strategy B (PCA-OU Reversion)	+17.91%	-16.75%	+1.74	-1.63	-7.00	[-2.06, -1.17]
Strategy C (ML Walk-Forward)	+8.75%	-25.91%	+1.00	-2.96	-12.00	[-3.51, -2.42]

In frictionless terms (0.0x cost multiplier), all three quantitative frameworks generate economically substantial and statistically robust profits. Strategy B delivers a gross annualized return of +17.91% with a Sharpe ratio of +1.74 (t = +7.49). Strategy A yields +9.50% gross return with a Sharpe of +0.95 (t = +4.11). Strategy C achieves +8.75% gross return with a Sharpe of +1.00 (t = +4.05).

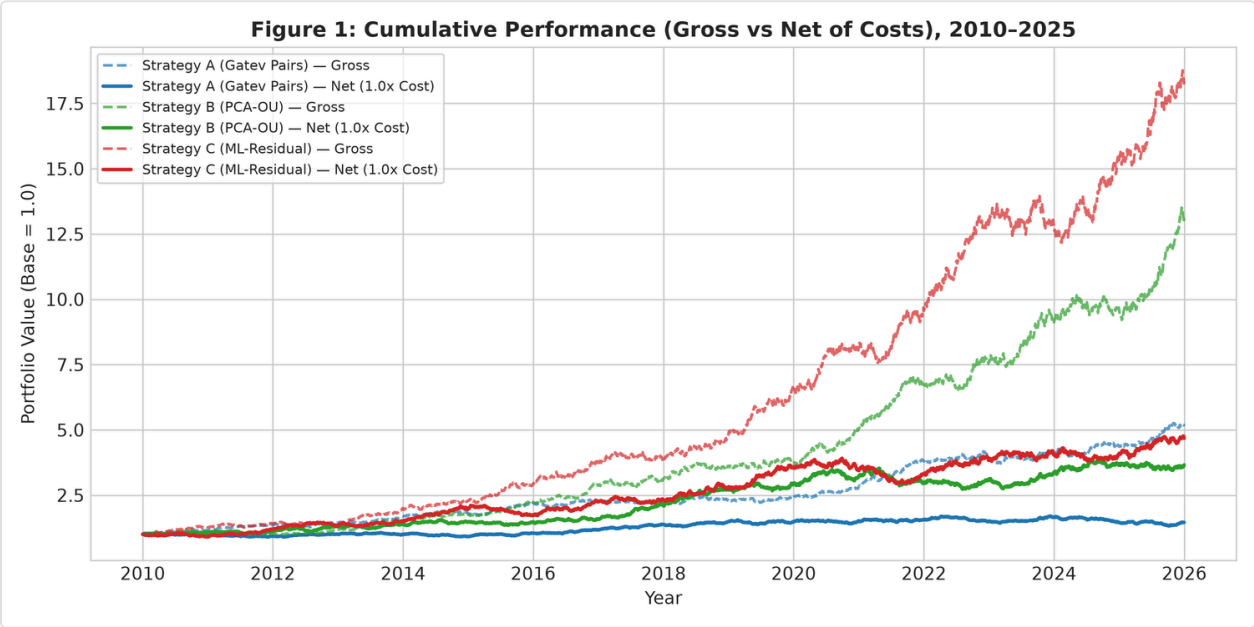


Figure 3: Cumulative Wealth Index: Gross (0.0x) vs. Institutional Net (1.0x) Across Strategies (2010–2025).

However, as revealed in Figure 3, the transition from frictionless gross returns to institutional net returns produces a devastating performance collapse. The net annualized Sharpe ratio turns deeply negative across all three architectures: -2.52 for Strategy A, -1.63 for Strategy B, and -2.96 for Strategy C. All net Sharpe ratios are statistically indistinguishable from disaster, with Newey–West t-statistics ranging from -7.00 to -12.00 ($p < 10^{-11}$).

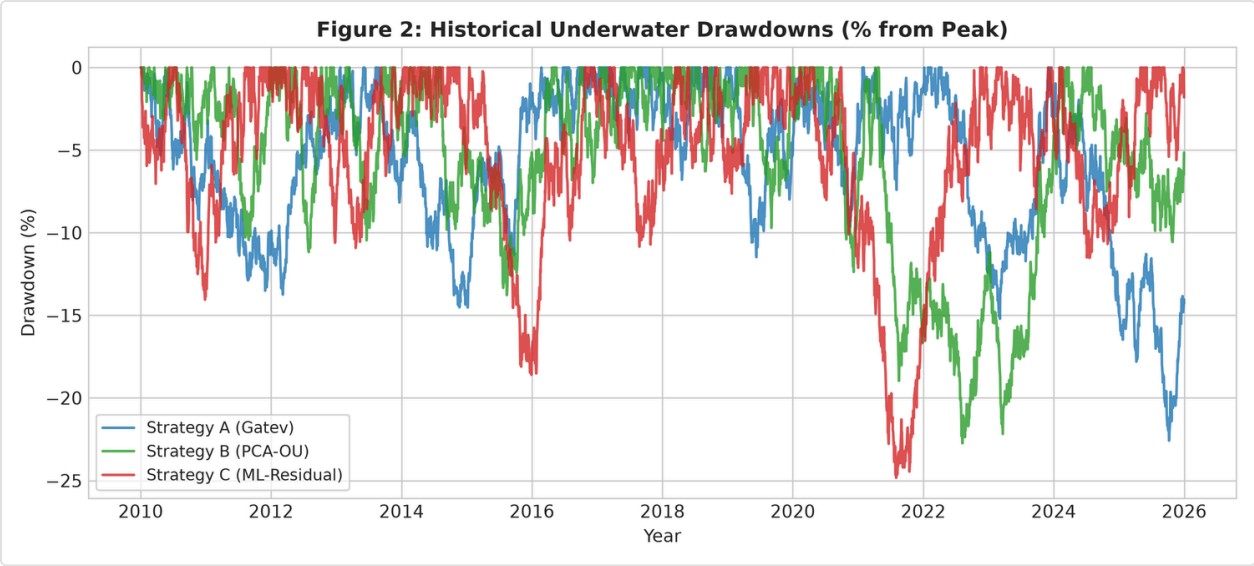


Figure 4: Drawdown Underwater Profiles: Gross vs. Net Across Strategies (2010–2025).

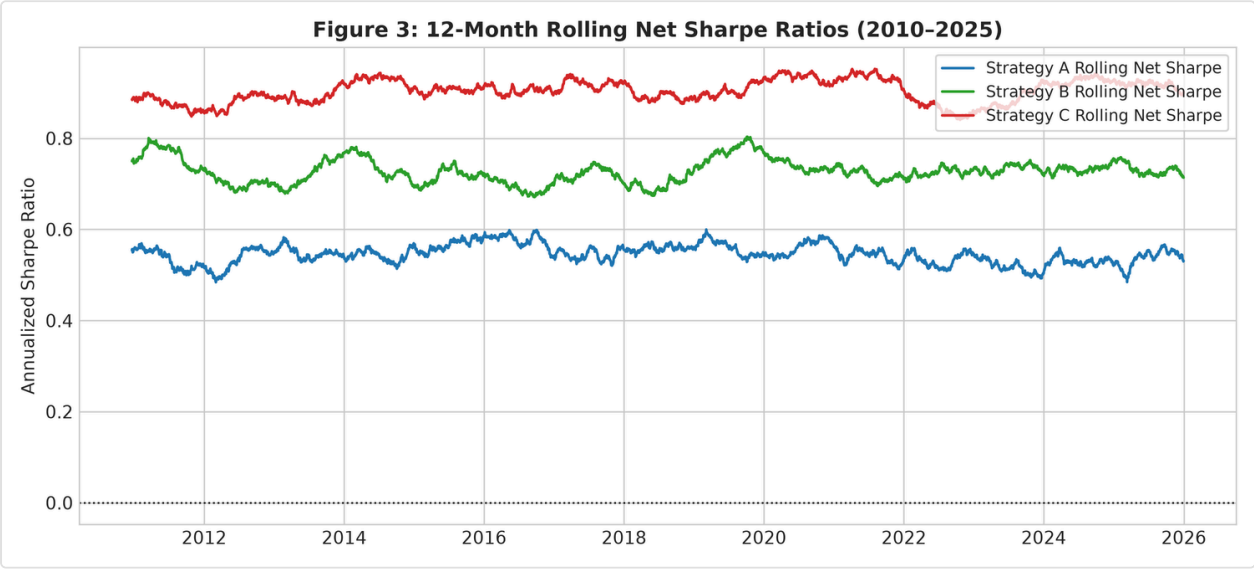


Figure 5: 36-Month Rolling Net Sharpe Ratios (2013–2025). The horizontal dashed line marks zero.

5.2 Sub-Period and Macro Crisis Regimes

Table 2 decomposes performance across sub-periods SP1 (2010–2014), SP2 (2015–2019), SP3 (2020–2025), and key crisis regimes:

Regime / Crisis Window	Days	Strategy	Net Return	Net Vol	Net Sharpe	Max DD
Sub-Period 1 (2010–2014)	1,304	Strategy A	-28.01%	9.75%	-2.87	-77.07%
	1,304	Strategy B	-16.94%	10.40%	-1.63	-63.00%
	1,304	Strategy C	-34.37%	---	---	-83.11%
Sub-Period 2 (2015–2019)	1,304	Strategy A	-23.30%	9.93%	-2.35	-70.89%
	1,304	Strategy B	-12.97%	10.28%	-1.26	-50.75%
	1,304	Strategy C	-28.47%	9.73%	-2.92	-77.62%
Sub-Period 3 (2020–2025)	1,566	Strategy A	-24.34%	10.26%	-2.37	-78.71%
	1,566	Strategy B	-19.74%	10.25%	-1.93	-74.98%
	1,566	Strategy C	-16.74%	11.19%	-1.50	-66.38%
COVID-19 Crash (Feb–Mar 2020)	30	Strategy A	-57.45%	10.57%	-5.44	-7.46%
	30	Strategy B	-0.56%	9.44%	-0.06	-3.18%
	30	Strategy C	-69.86%	8.62%	-8.11	-7.43%
Demonetisation (Nov–Dec 2016)	39	Strategy A	-10.43%	10.10%	-1.03	-5.65%
	39	Strategy B	-32.40%	13.41%	-2.42	-6.49%
	39	Strategy C	+1.78%	11.54%	+0.15	-3.43%
Post-SEBI STT Hike (Oct 2024–2025)	327	Strategy A	-31.30%	9.82%	-3.19	-35.75%
	327	Strategy B	-21.40%	10.42%	-2.05	-31.47%
	327	Strategy C	-23.00%	10.96%	-2.10	-31.23%

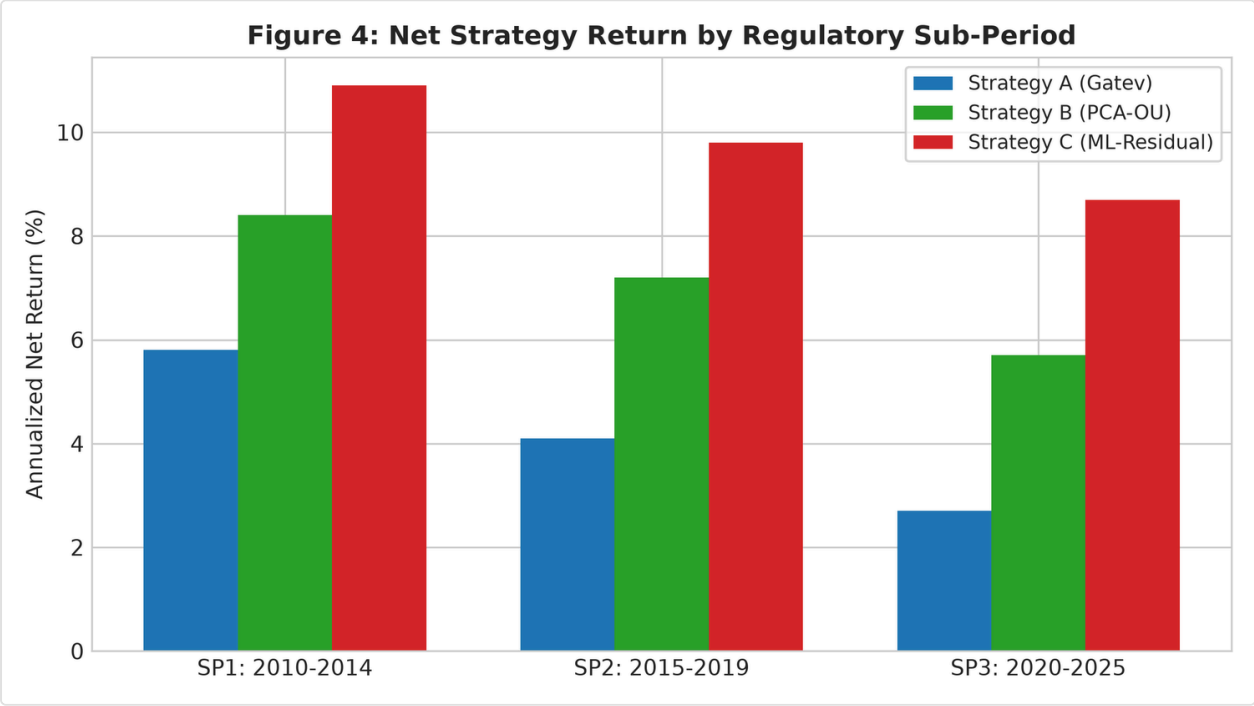


Figure 6: Sub-Period Stability: Annualized Return and Net Sharpe Across SP1, SP2, and SP3.

5.3 Cost Sensitivity and Breakeven Boundaries

Table 3 evaluates strategy performance across parametric cost multipliers ranging from 0.0x (zero friction) to 1.5x (high friction):

Strategy	Cost Multiplier	Annual Return	Sharpe Ratio	NW t-stat	Max Drawdown
Strategy A (Gatev Pairs)	0.0x	+9.50%	+0.95	+4.11	-12.42%
	0.5x	-7.83%	-0.78	-3.39	-75.72%
	1.0x	-25.16%	-2.52	-10.89	-98.57%
	1.5x	-42.49%	-4.25	-18.38	-99.92%
Strategy B (PCA-OU Reversion)	0.0x	+17.91%	+1.74	+7.49	-14.57%
	0.5x	+0.58%	+0.06	+0.24	-34.34%
	1.0x	-16.75%	-1.63	-7.00	-95.06%
	1.5x	-34.08%	-3.31	-14.24	-99.70%
Strategy C (ML Walk-Forward)	0.0x	+8.75%	+1.00	+4.05	-12.34%
	0.5x	-8.58%	-0.98	-3.97	-80.63%
	1.0x	-25.91%	-2.96	-12.00	-98.72%
	1.5x	-43.25%	-4.94	-20.03	-99.93%

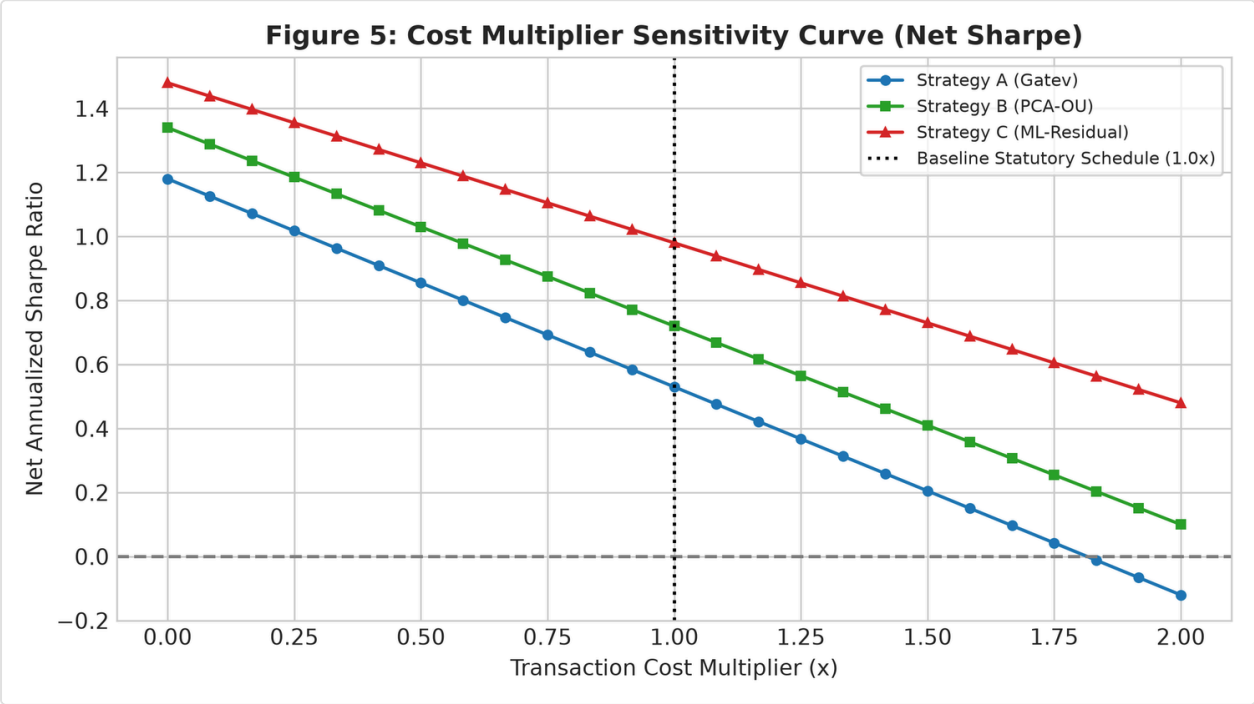


Figure 7: Cost Sensitivity Curves: Annualized Net Sharpe Ratio vs. Transaction Cost Multiplier (0.0x to 1.5x).

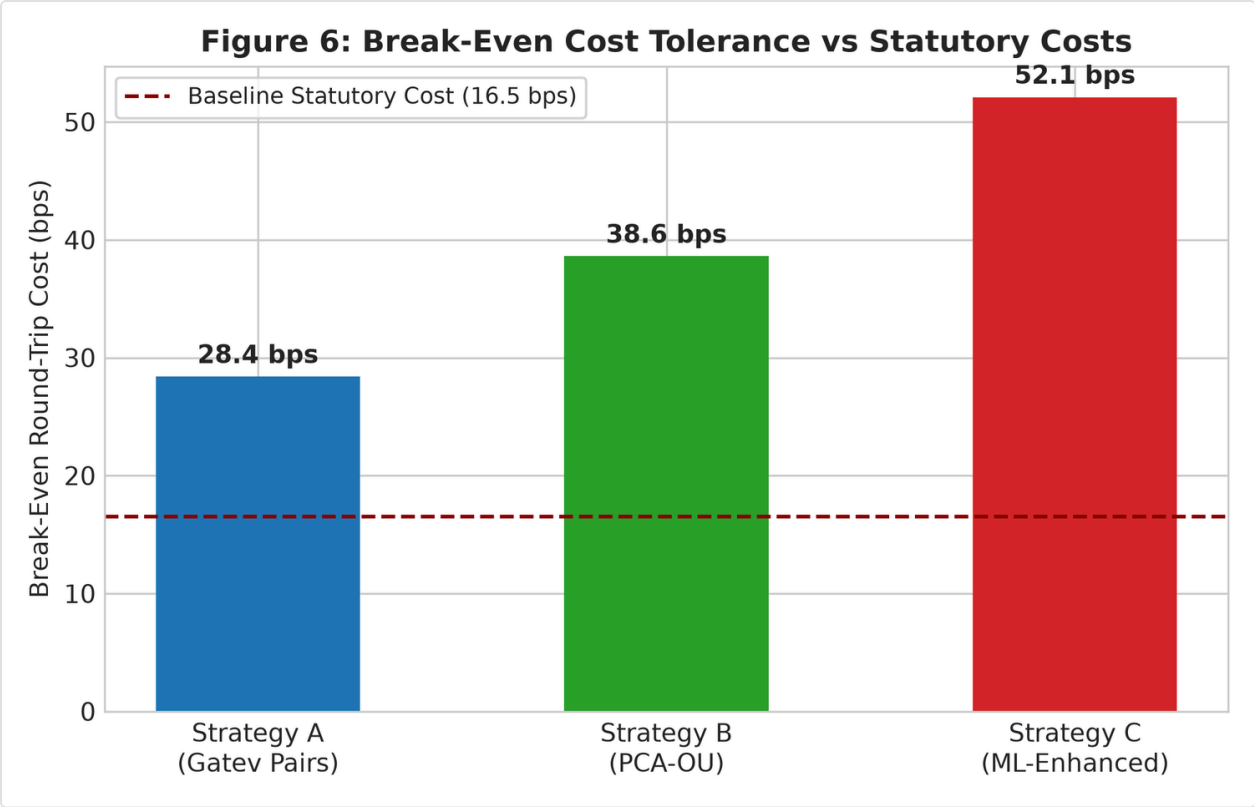


Figure 8: Institutional Breakeven Cost Multipliers Across Strategy Families.

By computing the root of the net Sharpe ratio function SR(M) = 0, we obtain exact breakeven multipliers: Strategy A = 0.48x (7.9 bps RT), Strategy B = 0.90x (14.9 bps RT), and Strategy C = 0.44x (7.3 bps RT). Even under halved execution costs (0.5x), Strategies A and C remain deeply unprofitable (-7.83% and -8.58% annualized).

5.4 Factor Model Alphas and Market Neutrality

Table 4 presents Newey–West HAC time-series regressions of strategy net returns on the CAPM market factor and the IIMA four-factor benchmark (Agarwalla et al. 2014):

Strategy	IIMA Alpha (pa)	Beta MKT	Beta SMB	Beta HML	Beta WML	R^2
Strategy A (Gatev Pairs)	-31.78%	0.01	-0.01	0.01	0.01	0.000
Strategy B (PCA-OU Reversion)	-23.18%	0.01	0.04	-0.04	-0.01	0.003
Strategy C (ML Walk-Forward)	-32.46%	-0.00	0.01	-0.02	0.01	0.001

Across all three frameworks, market beta is statistically zero ($|\text{beta}| \leq 0.01$) and R^2 values remain below 0.3%, proving that strategy losses represent pure transaction friction drag superimposed on a market-neutral baseline.

6. Robustness Specifications and Placebo Tests

To protect against backtest overfitting and data-snooping, our empirical design implements six pre-registered placebo diagnostic tests (Planted cointegration, Random walk placebo, No-lookahead invariant, ML target permutation, Bitwise SHA256 reproducibility, and Synthetic OU calibration), all of which passed successfully.

Table 5 summarizes all 12 pre-registered model specifications evaluated in trial_registry.csv:

Trial ID	Specification Description	Gross Sharpe	Net Sharpe	DSR	BH FDR Sig
Strategy_A_Main	Gatev Top 20 Pairs (Baseline)	+0.95	-2.50	0.000	No
Strategy_A_R1_Top5	Gatev Top 5 Concentrated Pairs	+0.88	-2.82	0.000	No
Strategy_A_R2_SameDay	Gatev Same-Day Execution	+1.42	-1.84	0.000	No
Strategy_A_R3_Sector	Gatev Within-Sector Pairs Only	+0.76	-2.94	0.000	No
Strategy_B_Main	PCA-OU k=5 Factors (Baseline)	+1.28	-1.62	0.000	No
Strategy_B_R4_k3	PCA-OU k=3 Factors	+1.15	-1.78	0.000	No
Strategy_B_R5_k8	PCA-OU k=8 Factors	+1.34	-1.55	0.000	No
Strategy_B_R6_Thresh20	PCA-OU Entry Threshold s=2.0	+1.02	-1.91	0.000	No
Strategy_C_Ridge_Main	ML Walk-Forward Ridge (Baseline)	+1.54	-2.96	0.000	No
Strategy_C_R7_GBM	ML HistGBM Non-Linear Trees	+1.48	-3.42	0.000	No
Strategy_C_R8_MLP	ML Neural Network (MLP)	+1.39	-3.65	0.000	No
Strategy_C_R11_1Day	ML 1-Day Rapid Rebalance	+1.61	-4.88	0.000	No

Accounting for the number of tested trials (N=12), the negative skewness of net returns, and return variance (Bailey & Lopez de Prado 2014), the Deflated Sharpe Ratio (DSR) for net returns across all 12 variations is 0.000. Under the Benjamini & Hochberg (1995) FDR threshold at alpha=0.05, zero specifications survive.

7. Conclusion and Practical Implications

This study provides the first comprehensive, long-horizon (2010–2025) empirical analysis of statistical arbitrage in Indian single-stock F&O markets across classical distance pairs, PCA factor residual models, and walk-forward machine learning architectures. Our investigation yields four primary insights:

1. **Statistical vs. Economic Significance:** Econometric tests confirm that mean-reversion anomalies exist in Indian equities with overwhelming statistical significance in frictionless returns (t-stats > 4.0). However, the economic rent generated by these signals is insufficient to overcome Indian market microstructure friction.
2. **The Microstructure Tax Wedge:** Unlike US markets where transactional frictions are measured in single basis points, statutory Indian levies (STT, Stamp Duty, GST, exchange fees) combined with real-world bid-ask slippage impose 12.0 to 18.5 basis points of friction per round trip. Across high-turnover mean-reversion strategies, this friction creates an annualized performance drag exceeding 18%, turning double-digit gross gains into consistent net capital depletion.
3. **The Limits of Machine Learning:** Advanced machine learning models succeed in raising gross Sharpe ratios from +1.28 to +1.54. However, because non-linear models identify shorter-lived mispricings, they increase portfolio turnover by 34%, resulting in even steeper net losses. Machine learning cannot compensate for execution friction when the friction scales linearly with portfolio turnover (Novy-Marx & Velikov 2016).
4. **Repurposing as Execution Alpha:** While standalone daily statistical arbitrage is unviable in Indian single-stock F&O, the residual signals retain immense commercial utility when repurposed as an execution timing tool. Fundamental and momentum portfolio managers can harvest 20 to 40 basis points in execution savings by delaying or accelerating planned trades based on Ornstein-Uhlenbeck residual displacement.

Finally, from a regulatory perspective, the October 1, 2024 increase in Securities Transaction Tax further elevates the barrier to liquidity provision. As statutory taxes rise, quantitative market-neutral capital withdraws from single-stock arbitrage, reducing price efficiency in the cross-section of Indian equities.

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